

Assignment 1

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if your idea is (mostly) correct and you haven't showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (25 marks) Let $F_1, \dots, F_p$ be p events. Suppose that $p = \text{poly}(\lambda)$ and for every $i \in [p]$ we have

$$\Pr[F_i] = \text{overwhelm}(\lambda).$$

Show that

$$\Pr[F_1 \wedge F_2 \wedge \dots \wedge F_p] = \text{overwhelm}(\lambda).$$

Problem 2 (25 marks) Show that public-key encryption IND-CPA security $\iff$ IND-mCPA security

Problem 3 (25 marks)

- Plain RSA Encryption Scheme:
 - Gen: Choose two large prime numbers P, Q and compute $N = P \times Q$. Choose a public exponent e such that $\gcd(e, \phi(N)) = 1$, where $\phi(N) = (P-1)(Q-1)$. Compute $d = e^{-1} \bmod \phi(N)$. The public key is (N, e) and the private key is (N, d) .
 - Enc: To encrypt a plaintext $M \in \mathbb{Z}_N^*$, compute $C = M^e \bmod N$.
 - Dec: To decrypt a ciphertext C , compute $M = C^d \bmod N$.

Prove that Plain RSA Encryption Scheme is not IND-CPA secure.

Problem 4 (25 marks) Consider the following public-key encryption scheme. The public key is (G, q, g, h) and the private key is x , generated exactly as in the El Gamal encryption scheme. In order to encrypt a bit b , the sender does the following:

1. If $b = 0$ then choose a uniform $y \in \mathbb{Z}_q$ and compute $c_1 := g^y$ and $c_2 := h^y$. The ciphertext is (c_1, c_2) .
2. If $b = 1$ then choose independent uniform $y, z \in \mathbb{Z}_q$, compute $c_1 := g^y$ and $c_2 := g^z$, and set the ciphertext equal to (c_1, c_2) .

Show that it is possible to decrypt efficiently given knowledge of x . Prove that this encryption scheme is CPA-secure if the decisional Diffie Hellman problem is hard.